← BACK ~

● ERROR 404 // PATH_RESOLUTION_FAILED

404

LOST IN *the void.*

The coordinates you've requested do not exist within this namespace. The system has reached a terminal boundary.

0x0EFA

← STEP BACK

● SYSTEM.READY ● CACHE.PURGED

← BACK ~

JUSTINCLARKE@PORTFOLIO: ~/CONNECT

email linkedin github | Royel Otis - Linger - SiriusXM Session | ● shipping

~$ connect ▌

justin *clarke.*

data & analytics developer.

└ stack: python sql ● dbt power-bi

↳ Data to decisions and decisions into products

● AVAILABLE   VISA · SPONSORED

SAVE CONTACT   BOOK CALL   QR CODE

⏎ or press C to copy email

// CHANNELS · 05   sorted by: response_time ↓

[01] EMAIL — justinsavioclarke@outlook.com — REPLIES < 24H →

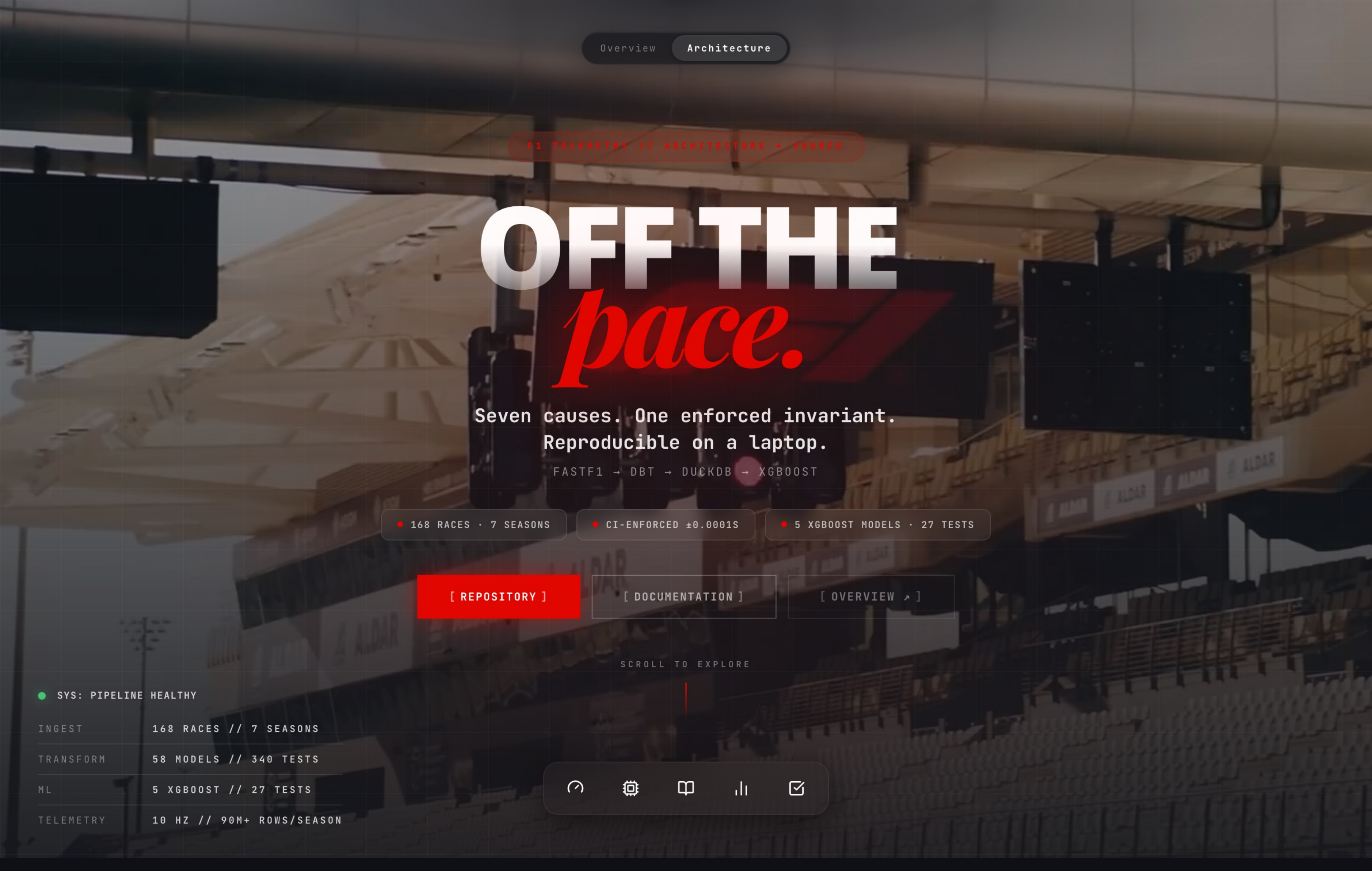
[02] LINKEDIN — /in/justinsavioclarke — PROFESSIONAL NETWORK →

[03] GITHUB — /justinclarke — 38 PUBLIC REPOS →

[04] BOOK — cal.com/justinclarke — THIS WEEK: OPEN →

[05] INQUIRIES — let's work together — SECURE FORM →

Overview Architecture



F1 TELEMETRY // ARCHITECTURE + SOURCE

OFF THE *pace.*

Seven causes. One enforced invariant.
Reproducible on a laptop.

FASTF1 → DBT → DUCKDB → XGBOOST

168 RACES · 7 SEASONS | CI-ENFORCED ±0.0001S | 5 XGBOOST MODELS · 27 TESTS

[REPOSITORY] [DOCUMENTATION] [OVERVIEW ↗]

SCROLL TO EXPLORE

● SYS: PIPELINE HEALTHY

INGEST	168 RACES // 7 SEASONS
TRANSFORM	58 MODELS // 340 TESTS
ML	5 XGBOOST // 27 TESTS
TELEMETRY	10 HZ // 90M+ ROWS/SEASON

01 / CAPABILITIES & PROBLEMS SOLVED

WHAT THIS SYSTEM DOES. WHAT MAKES IT HARD.

A headless physics pipeline producing ML-ready feature marts from raw 10Hz F1 telemetry. Three causal identification problems each with an explicit engineering solution baked into the codebase.

● ACTIVE IDENTIFICATION PROBLEMS

01

ACTIVE

ERR_CONFOUNDER_BIAS

SIMULTANEOUS CONFOUNDERS

Tyre age, fuel load, and traffic change at the same time naïve regression models confound them all.

● ✓ RESOLUTION

> Calibrated teammate-panel controls that isolate tyre wear from weight change and constructor pace.

02

ACTIVE

ERR_CIRCULAR_WAKE

DIRTY-AIR ENDOGENEITY

Relative pace determines track position, creating circular causality with the trailing-car wake tax.

● ✓ RESOLUTION

> Lagged instruments (Frisch-Waugh-Lovell) isolating wake tax without circular causation.

03

ACTIVE

ERR_DRIFT_VIOLATION

ADDITIVE INVARIANCE

The seven components must reconstruct the observed lap time exactly any drift poisons downstream ML.

● ✓ RESOLUTION

> CI-enforced singular test assert_lap_7term_identity fails the build if identity breaks.

● FEATURE CAPABILITIES

GHOST CAR

◆ PLANNED

Counterfactual lap reconstruction recombines driver skill residuals with any constructor baseline.

FCT_GHOST_CAR_PACE PLANNED

OVERTAKE GRAPH

● BUILT

On-track pass detection via driver_ahead_id swaps. Filters pit-cycle gains, SC restarts, and lapping traffic.

INT_OVERTAKES FCT_RACECRAFT

TYRE CLIFF PREDICTOR

● BUILT

Multi-horizon XGBoost targets for degradation jumps. 5 models, 38 features, 110K training laps.

FCT_CLIFF_PREDICTION XGBOOST

POWERTRAIN FINGERPRINT

● BUILT

Per-lap driving style signature gear changes, RPM profiles, throttle %, DRS share, short-shift index.

INT_LAP_POWERTRAIN

PIT STRATEGY VALUE

● BUILT

Classifies each pit call as optimal, overran, undercut_forced, or early.

INT_PIT_STRATEGY_VALUE FCT_STINT

RACE CONTROL

● BUILT

SC/VSC windows, red-flag laps, and penalties parsed with precise lap-level neutralisation.

STG_RACE_CONTROL INT_EVENTS

02 / INGESTION : BRONZE LAYER

168 RACES INGESTED. 4 DATASETS PER RACE.

FastF1 + OpenF1 → Hive-partitioned Parquet, organised by `season/race_id/dataset`. JSON-Schema contracts validated at ingest. `data_quality.py` gates reject under-lap counts and telemetry gaps before any transform runs.

LAPS	TELEMETRY	WEATHER	RACE CONTROL
Timing, sector splits, tyre data, lap flags	10Hz positional + RPM/gear/throttle/DRS/X/Y/Z	Air/track temp, humidity, barometric pressure	SC/VSC windows, penalties, flags, messages

INGESTION PIPELINE

FastF1 → OpenF1 → JSON-Schema validation → data_quality.py gates → Hive-partitioned Parquet

Partition key: season / race_id / dataset. Gates reject under-40-lap races and telemetry samples below 8Hz before they reach Silver. 168 races ingested; 147 fully processed in fct_lap_residuals.

168	147	90M+	10Hz
RACES INGESTED	RACES PROCESSED	TELEMETRY ROWS	TELEMETRY RATE
BRONZE COVERAGE : 168	IN FCT_LAP_RESIDUALS	PER SEASON	POSITIONAL DATA

03 / TRANSFORM : DAG & INVARIANT

58 DBT MODELS. ONE ENFORCED INVARIANT.

Every model is a node in a directed acyclic graph. Every edge has a test. The 7-term identity - components sum to within ±0.0001s of the actual lap time - runs on CI against every race in the dataset. If it breaks, the build breaks.

LAYER 0 // BRONZE

INGESTION

FastF1 + OpenF1 → Hive-partitioned Parquet. 10Hz positional telemetry, race control messages, pit data, lap times.

FASTF1 OPENF1 PARQUET HIVE

LAYER 1 // SILVER

STAGING + PHYSICS

36 intermediate models. Fuel state, stint geometry, dirty-air OLS (lagged instrument), tyre cliff prediction.

DBT-CORE DUCKDB OLS FRISCH-WAUGH

LAYER 2 // GOLD

FEATURE MARTS

9 production mart tables. fct_lap_residuals, fct_driver_skill_features, fct_cliff_prediction_features, fct_ghost_car_pace, fct_racecraft.

ML-READY 1-LAP GRAIN 340 TESTS

LAYER 3 // MODELS

ML TARGETS

Five XGBoost models - degradation quantiles (p10/p50/p90), cliff classifier, and stint-life regressor. All trained on fct_cliff_prediction_features, ONNX-exported. Ghost Car app - planned.

XGBOOST 5 MODELS ONNX

STACK

- → **dbt-core** + DuckDB (local)
- → **FastF1** Python Ingestion
- → **GitHub Actions** CI
- → **React + DuckDB-Wasm** (planned)
- → **XGBoost** (built)

TEST COVERAGE

340

schema + singular + invariant tests. 100% pass rate across all 147 races, every build.

● ALL PASSING

TELEMETRY RESOLUTION

10 Hz

Raw positional, RPM, gear, throttle, DRS state, X/Y/Z coordinates from FastF1. ~90M rows per season.

MODEL LINEAGE (SIMPLIFIED DAG PATH)

stg_laps ⟶ int_stint_geometry ⟶ int_lap_fuel_state
stg_telemetry ⟶ int_lap_air_state ⟶ int_dirty_air_tax_component
stg_race_control ⟶ int_race_control_events ⟶ int_event_corrections
stg_weather ⟶ int_track_evolution ⟶ int_constructor_structural_pace
⬇ converge INT_LAP_RESIDUAL_DECOMPOSED ⟶ FCT_LAP_RESIDUALS

BUILD TIME

<4 SEC

Full DAG build locally on DuckDB. Zero cloud warehouse. Zero cloud cost.

● MAKE DBT-DEV

02 / ARCHITECTURE & SCALE

ZERO CLOUD COMPUTE.

Built a headless dbt pipeline running entirely locally on DuckDB. 58 models compile and run 340 tests in under 4 seconds. Maximum developer velocity, rigorous CI/CD constraints, and zero cloud costs.

ASSERT_ADDITIVE_IDENTITY.SQL · CI GATE ● CI ENFORCED

```
-- assert_lap_7term_identity  - fails build if 7-term sum ≠ pace_delta_s ± 0.

SELECT count(*) FROM fct_lap_residuals
WHERE abs(pace_delta_s - (
  fuel_component_s + compound_component_s +
  rubber_component_s + ambient_component_s +
  constructor_component_s + dirty_air_tax_s +
  driver_skill_residual_s
)) > 0.0001

✓ Finished running 340 tests in 3.82s : PASS
✓ assert_lap_7term_identity ... PASS
```

03 / THE SCIENCE

ECONOMETRICS MEETS 10HZ TELEMETRY.

This isn't rolling averages. Isolating confounded variables from time-series race data demands genuine causal identification. We use lagged instruments, Frisch-Waugh detrending, and OLS estimation - not correlation.

CLICK ANY TERM TO EXPAND ITS ESTIMATION METHOD

$$\Delta LapTime = \theta_{fuel} \cdot F_i + \theta_{compound}(age_i) + \theta_{rubber} \cdot R_i + \theta_{ambient} \cdot W_i + \theta_{car} \cdot V_i + \hat{\theta}_{air} \cdot A_{i-1} + \varepsilon_i \text{ (driver)}$$

θ̂_AIR : DIRTY AIR TAX (CAUSAL OLS WITH LAGGED INSTRUMENT)

SQL / HEURISTIC MAPPING: `dirty_air_share_lag1 * dirty_air_coefficient_s`

The pace cost of following another car in turbulent air. Confounders abound - a driver trailing closely is often faster (DRS boost) or slower (aero degradation) at the same time. We use a 1-lap lag instrument - dirty air share from the previous lap - to ensure causal ordering. Lapping-traffic laps are excluded from the calibration panel to prevent bias. Global θ̂_air estimated via SQL OLS: COVAR_POP(residual, air_lag1) / VAR_POP(air_lag1). Bounded [0, 5.0s] per lap.

05 / THE OUTPUT

A CAUSAL DECOMPOSITION, DOWN TO THE MILLISECOND.

The engine doesn't output a chart. It outputs a pristine, 1-lap grain feature matrix : **`fct_lap_residuals`** - where each component is a precisely isolated, physically-grounded additive term in seconds. Seven terms. CI-enforced identity. Positive = slower.

SEE THE OUTPUT →

FCT_LAP_RESIDUALS · DUCKDB · DATA/DEV.DUCKDB ● 340 TESTS PASSING

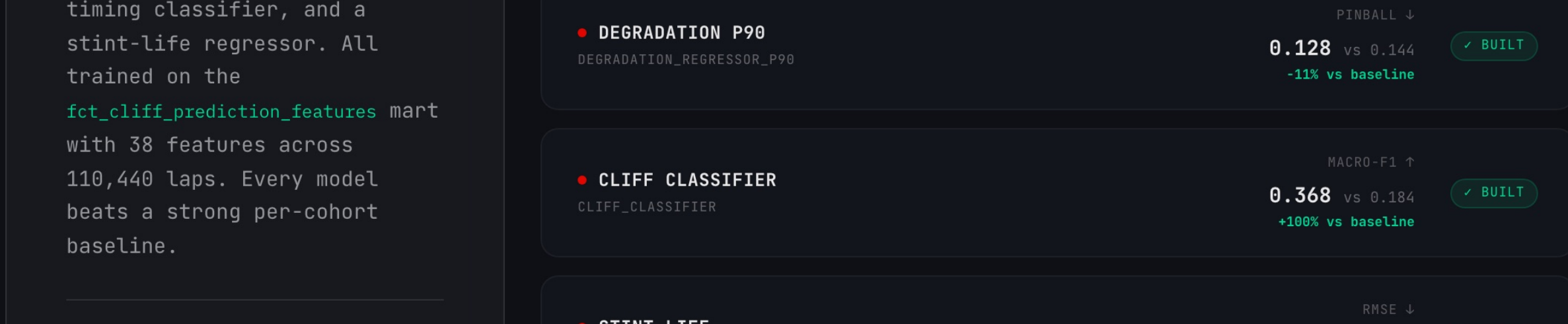
```
$ duckdb data/dev.duckdb
SELECT lap_id, driver_id, lap_time_s,
       fuel_component_s, compound_component_s, rubber_component_s,
       ambient_component_s, constructor_component_s,
       dirty_air_tax_s,
       driver_skill_residual_s,  -- the human element
       ml_eligible
FROM fct_lap_residuals
WHERE ml_eligible = TRUE
  AND race_year = 2021 AND race_id = '2021_22'
ORDER BY driver_id, lap_number LIMIT 5;
```

VIEW SAMPLE: 2021 ABU DHABI ⌄ ● LIVE RESULT

DRIVER_ID	LAP_TIME_S	FUEL_COMP_S	COMPOUND_S	RUBBER_S	AMBIENT_S	CONSTRUCTOR_S	DIRTY_AIR_S	DRIVER_SKILL_S	ML_ELIGIBLE
HAM	88.392	+1.241	+0.762	+0.214	+0.082	-0.310	0.000	-0.184	TRUE
VER	88.651	+1.239	+0.780	+0.198	+0.081	+0.112	+0.218	+0.074	TRUE
BOT	88.910	+1.237	+0.771	+0.221	+0.083	-0.309	+0.401	+0.127	TRUE
PER	88.820	+1.238	+0.778	+0.205	+0.082	+0.109	0.000	+0.097	TRUE
LEC	89.441	+1.235	+0.795	+0.289	+0.083	+0.387	+0.102	+0.131	TRUE

LAP DECOMPOSITION · 2021 ABU DHABI · LAP 14

Same lap, decomposed - the red SKILL segment is the only part the driver controls.



06 / MACHINE LEARNING

FIVE XGBOOST MODELS.

A quantile trio (p10/p50/p90) for degradation pace, a cliff-timing classifier, and a stint-life regressor. All trained on the `fct_cliff_prediction_features` mart with 38 features across 110,440 laps. Every model beats a strong per-cohort baseline.

Calibration (p10–p90)	0.804 empirical (18,677 laps)
Mean interval width	1.11s
ONNX parity	all 5 boosters · atol=1e-5
Leakage probe (race_year)	0.9999 → excluded

Model	Metric	Score	Change	Status
● **DEGRADATION P10** DEGRADATION_REGRESSOR_P10	PINBALL ↓	0.078 vs 0.128	-39% vs baseline	✓ BUILT
● **DEGRADATION P50** DEGRADATION_REGRESSOR_P50	PINBALL ↓	0.190 vs 0.225	-16% vs baseline	✓ BUILT
● **DEGRADATION P90** DEGRADATION_REGRESSOR_P90	PINBALL ↓	0.128 vs 0.144	-11% vs baseline	✓ BUILT
● **CLIFF CLASSIFIER** CLIFF_CLASSIFIER	MACRO-F1 ↑	0.368 vs 0.184	+100% vs baseline	✓ BUILT
● **STINT LIFE** STINT_LIFE_REGRESSOR	RMSE ↓	7.73 vs 8.08	-4% vs baseline	✓ BUILT

07 / VALIDATION & HONESTY

TRAINED 2018–2024. 2025 IS THE HOLDOUT.

The 2025 season is a designated reproducible out-of-sample holdout. Until it ingests, headline numbers come from the final TimeSeriesSplit fold (2024). The switch to a true holdout requires no code change.

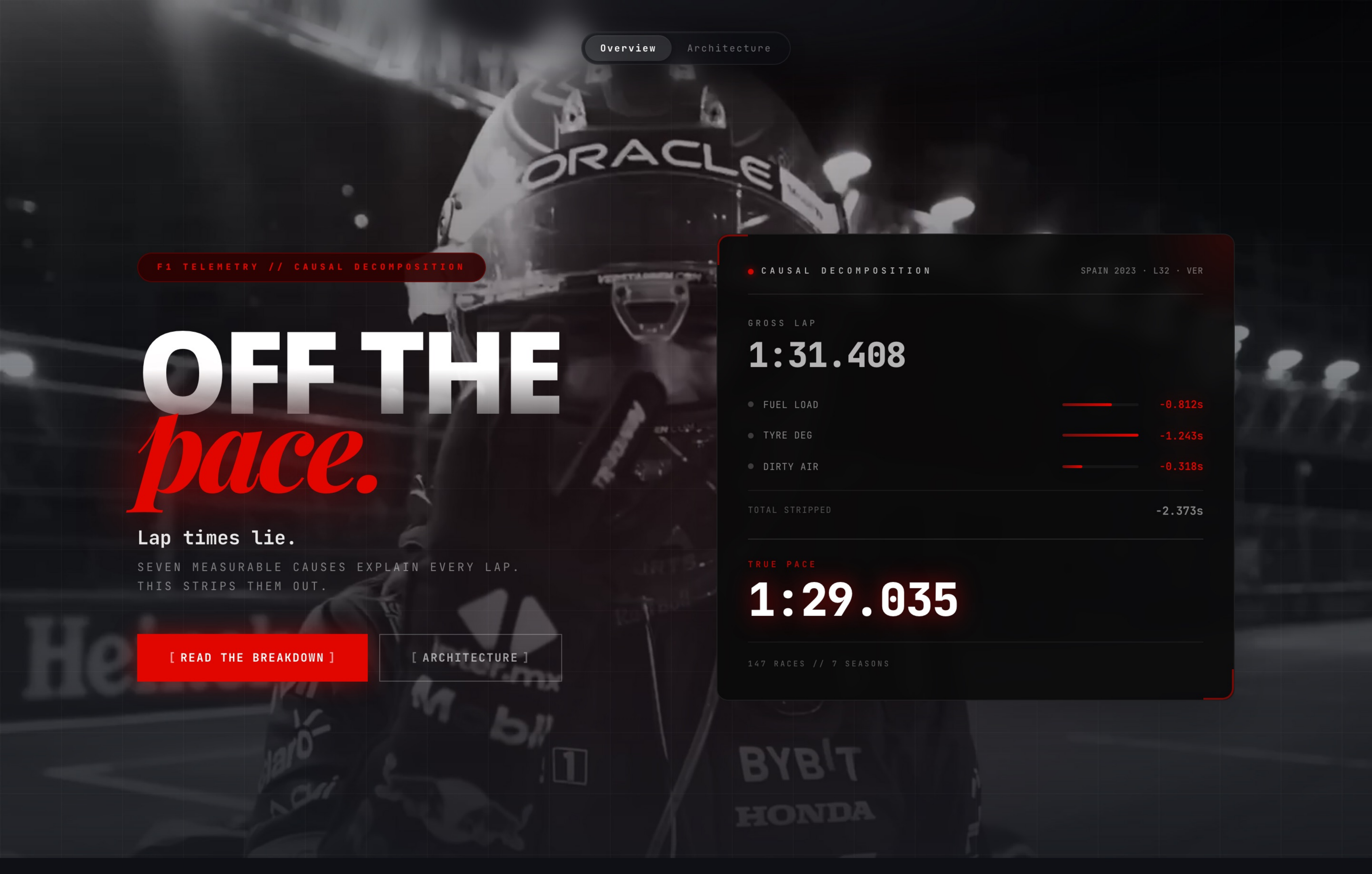
MAKE TEST-ALL · CI GATE

```
$ make test-all
cd transform && dbt build --profiles-dir profiles --
target ci
```

PROJECT STATUS - HONEST BUILD VS PLANNED

Ingestion (Bronze)	✓ BUILT
Transform (58 models, 340 tests)	✓ BUILT
Coefficients - all 6 components fitted	✓ FITTED

Overview Architecture



F1 TELEMETRY // CAUSAL DECOMPOSITION

OFF THE *pace.*

Lap times lie.

SEVEN MEASURABLE CAUSES EXPLAIN EVERY LAP.
THIS STRIPS THEM OUT.

[READ THE BREAKDOWN] [ARCHITECTURE]

CAUSAL DECOMPOSITION SPAIN 2023 · L32 · VER

GROSS LAP
1:31.408

FUEL LOAD	-0.812s
TYRE DEG	-1.243s
DIRTY AIR	-0.318s
TOTAL STRIPPED	-2.373s

TRUE PACE
1:29.035

147 RACES // 7 SEASONS

IN F1, THE STOPWATCH IS THE WORST LIAR IN THE PADDOCK.

Every lap time hides seven physically-grounded causes. This engine separates them:causally, additively, and to the millisecond:to isolate true performance.

CAR BASELINE · FUEL MASS · TYRE CHEMISTRY · RUBBER EVOLUTION · AIR DENSITY · DIRTY-AIR WAKE · DRIVER SKILL

0	0M+	0	0
RACES ANALYZED	TELEMETRY ROWS	DATA MODELS	PASSING TESTS

7 seasons · 10Hz telemetry · modeled locally · zero cloud spend

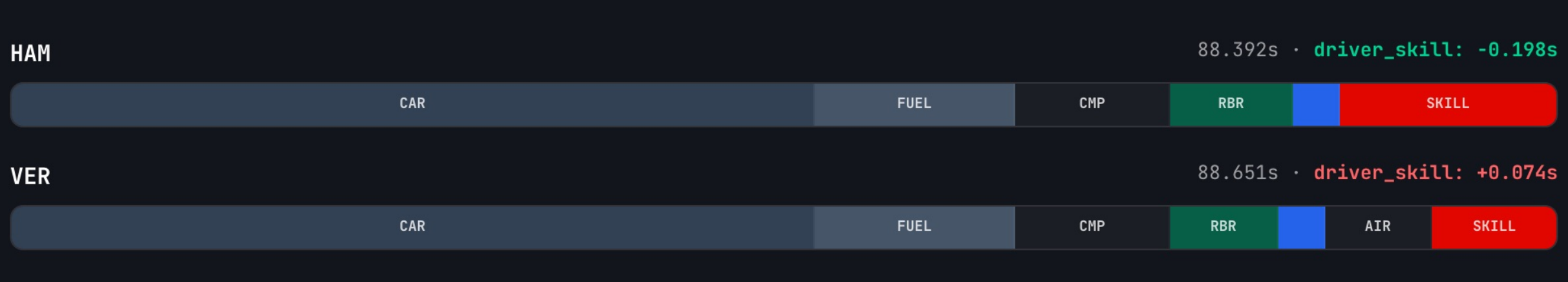
TECHNOLOGY STACK

Python	INGESTION 01	FastF1	TELEMETRY 02	dbt	TRANSFORM 03
SQL	QUERY 04	DuckDB	WAREHOUSE 05	XGBoost	ML 06
ONNX	INFERENCE 07	React	APP 08	GitHub Actions	CI 09

THE DECOMPOSITION

SAME LAP. SEVEN CAUSES.

2021 Abu Dhabi Grand Prix · Lap 14 · fct_lap_residuals



The red SKILL segment is the only part the driver controls. Everything else is physics - and now it's measured. Verstappen's dirty-air tax (AIR) cost him **+0.218s** in clean-air equivalent pace on this lap. The stopwatch lumped it into his split.

● Car baseline ● Fuel weight Tyre compound ● Track evolution ● Weather Dirty air ● Driver residual

ARCHITECTURE

THE DATA PIPELINE.

A headless physics engine transforming raw telemetry into clean, causal features.

01 / INGEST

FastF1 Telemetry

147 races × 4 datasets - laps, weather, race control, and 10Hz car telemetry - extracted locally into Hive-partitioned Parquet.

02 / MODEL

dbt DAG

58 models in DuckDB applying physics invariants to isolate car performance from tyre degradation, traffic, and fuel load.

03 / OUTPUT

Causal Predictors

Clean, counterfactual pace metrics ready for XGBoost models and simulation engines.

SCHEMA // MART/FCT_LAP_RESIDUALS.SQL ● COMPILE_SUCCESS

The pipeline compiles into `fct_lap_residuals` - a 1-lap-grain partition matrix where each of the 8 additive telemetry residuals is independently verified and CI-guarded.

Downstream consumers: XGBoost Regression, Survival Modeling, Monte Carlo Race Simulator.

COLUMN	TYPE
lap_id	VARCHAR (PK)
residual_skill	FLOAT64
residual_dirty_air	FLOAT64
residual_tyre_deg	FLOAT64

COMPLEXITY

THREE ENGINEERING PROBLEMS

Known failure modes in sports analytics, each with an explicit solution in this codebase.

ERR_CONFOUNDER_BIAS ● ACTIVE_WARN

SIMULTANEOUS CONFOUNDERS

Tyre age, fuel load, and traffic change at the same time, biasing simple regression models.

● ✓ SYSTEM RESOLUTION

› Calibrated teammate panel controls that isolate tyre wear from weight change.

ERR_CIRCULAR_WAKE ● ACTIVE_WARN

DIRTY-AIR ENDOGENEITY

Relative pace determines track position, creating circular causality with trailing wake.

● ✓ SYSTEM RESOLUTION

› Lagged instruments isolating wake tax via the Frisch-Waugh-Lovell theorem.

ERR_DRIFT_VIOLATION ● ACTIVE_WARN

ADDITIVE INVARIANCE

Model components must sum exactly to the observed lap time without drift.

● ✓ SYSTEM RESOLUTION

› Automated CI test assertion validating zero-drift constraints per lap row.

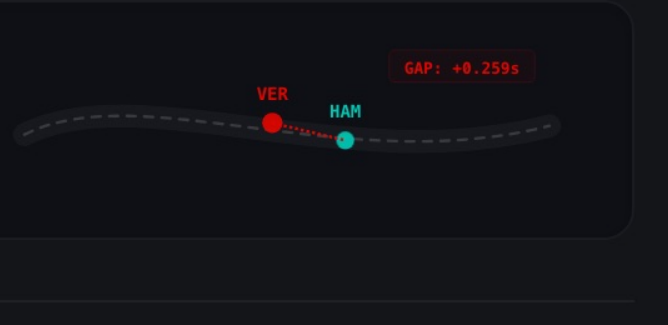
BUSINESS VALUE

WHY IT MATTERS.

Translating raw physics into strategic racing advantage and actionable insights.

Ghost Car : Planned

Application layer that reassembles micro-sectors from different drivers at identical track conditions - recombining driver skill residuals with any constructor baseline.



PLANNED - WHAT-IF PACE CEILING FOR ANY CAR+DRIVER PAIRING

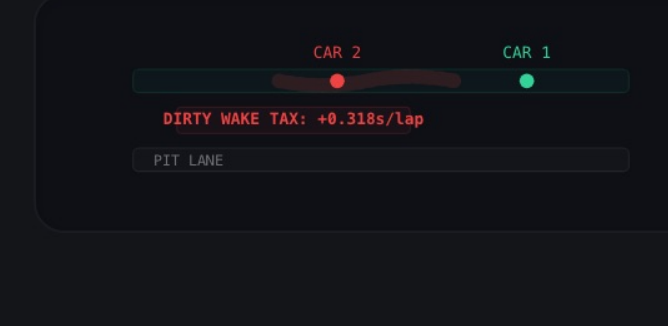
Tyre Cliff Prediction

Isolates purely mechanical grip degradation from fuel burn using survival modelling.

PREDICTS COMPOUND FAILURE CLIFF BEFORE IT HAPPENS

Pit Strategy Value

Quantifies the exact cost of dirty-air traffic using counterfactual clean-air times.



CALCULATES OVERCUT/UNDERCUT VIABILITY IN TRAFFIC

METHODOLOGY

RIGOUR BY DESIGN.

The architecture is built around the assumption that someone will try to break it.

● PASS

Causal identification

Frisch-Waugh-Lovell partialling + lagged instruments for endogenous dirty-air. Each term has a causal interpretation, not just a statistical fit.

● PASS

340 CI invariants

Every model is guarded by a singular SQL assertion. The additive-identity test fires if any term drifts by > 0.0001s. Build fails before bad data reaches ML.

● PASS

Reproducible locally

DuckDB on-disk warehouse, FastF1 via pip, dbt-core - no cloud account required. The full pipeline runs on a laptop. Zero vendor lock-in.

● PASS

Fully documented

Auto-generated reference docs from dbt manifests. Every model, column, and test is described and versioned. Docs can't drift from the code.

READ THE DOCS →

FEATURED INSIGHT

THE DATA SPEAKS.

Lap 14, 2021 Abu Dhabi Grand Prix. Hamilton and Verstappen on identical tyres, same fuel load, same weather. The engine disaggregates the 0.259s gap into its causes. Verstappen's raw pace was faster than his lap time showed - the dirty-air tax swallowed +0.218s of it.

TIMING_FEED // DRIVER_SKILL_RESIDUAL_S NEGATIVE = FASTER THAN EXPECTED

POS	DRIVER	SKILL RESIDUAL	RELATIVE PERFORMANCE	SECTOR NOTE
1	HAM	-0.198s		clean air, Mercedes
2	LEC	-0.131s		clean air, Ferrari
3	BOT	-0.104s		clean air, Mercedes
4	PER	-0.079s		no dirty-air penalty
5	VER	+0.043s		+0.218s dirty-air tax

Hamilton's advantage was real. Verstappen's dirty-air disadvantage was real. Without the decomposition, neither is visible in the timing sheets.

SEE FCT_LAP_RESIDUALS →

HONEST BUILD STATUS

This project is built layer-by-layer. Planned items are designed and specced, not yet shipped.

Ingestion (Bronze)	✓ BUILT	Transform (58 models, 340 tests)	✓ BUILT
Coefficients - all 6 components fitted	✓ FITTED	ML (5 XGBoost models, 27 tests)	✓ BUILT
Docs & Governance (Docusaurus + 3 CI)	✓ BUILT	First Attributed Finding	◇ IN PROGRESS
Frontend (React + DuckDB-Wasm)	○ PLANNED	Streaming	○ PLANNED

NEXT STEPS

SEEN THE PITCH. NOW SEE THE PROOF.

The Architecture view walks through the engineering - the DAG architecture, the CI gates, the science behind the seven-term identity, and the full validation ledger.

REPO DOCS VIEW THE ARCHITECTURE

Built by **Justin Clarke** | Licensed **AGPL-3.0** OFF THE PACE F1 Causal Pace Engine 2018–2024 ● PIPELINE HEALTHY

CONTACT ● LIVE

If your data isn't driving decisions, it's just *noise.*

GET IN TOUCH ↗

justinsavioclarke@outlook.com

← BACK ~

● AVAILABLE · DUBAI

VOL. I · NO. 01 *the long version* THU · 04 JUN 2026

by Justin Clarke

the long version.

What's left when the work is done. A short paper on noticing things, building when bored, and why one obsession leads to another.

§ I *Origin* p. 02
how curiosity became a job title.
§ II *Threads* p. 03
one thing tends to lead to another.
§ III *The Pattern* p. 05
an essay on noticing, written from a café.
§ IV *Off the Clock* p. 07
what fills the rest of the day.
§ V *A Few Things I Had to Learn* p. 09
things I picked up slowly, usually the hard way.
§ VI *Working Together* p. 11
if any of this landed, here is the door.

§ I

Origin.

- how curiosity became a job title -

Before I ever touched a dataset, my brother JAYDEN and I took apart a song. *The Sound of Silence*, the Central Park recording. We'd listen and strip it down by ear, one layer at a time. First the crowd goes. Then the bass. Then the acoustic guitar, then the harmony, until you're left with a single voice holding the whole structure up on its own. Once you hear music that way, you can't stop. Every song becomes a system of interlocking parts, and you start asking what each one is actually doing there.

That instinct never left. It only changed mediums. It turned a million Spotify playlists into a recommendation engine. It turned a Formula 1 race into a causal model that pulls a single lap apart to ask how much was the driver and how much was the car. It's the same move every time: take the thing apart, work out what each piece contributes, then rebuild the whole picture so someone else can see it too.

So no, I didn't start in data. I started in curiosity, and data is just where it eventually found a job. The degree was Computer Science. The MSc was a Distinction. The MBA is ongoing; weekends and evenings, between projects, between rooms. None of that is why I ended up here. I ended up here because I kept asking why the numbers weren't being used, and what it would take to actually use them. Nobody had a good answer. So I started building one.

I'm based in Dubai now. I build analytics systems on Microsoft Fabric, ship full-stack products, design interfaces, take photographs, and draw things by hand. These look like separate hobbies. They aren't. They're the same instinct rearranged for different rooms: *find the signal, make it legible, make it look like it deserves to be seen.*

The London years were where the breadth turned into a habit. Three of them. One at Queen Mary, two more figuring out what to do with what Queen Mary had given me. Somewhere in those three years, work-mode and rest-mode stopped being two different people and became one person with two registers. This issue is mostly about the second register, since the first one has its own website.

EDITOR'S NOTE
Justin lived in London 2022 – 2025. He cycles everywhere. He still misses the rain about thirty percent of the time. The other seventy percent, he doesn't.

§ II

Threads.

- one thing tends to lead to another -

Most of the work I'm proud of started somewhere else. In a hobby, a question I couldn't drop, a song I kept replaying past the point of reason. The pattern repeats often enough that it's worth naming. Here is what led to what.

John Grisham, junior high → *reading people.*

I read every courtroom thriller I could get my hands on before I was fourteen. Grisham teaches you something he never meant to: the detail that explains the ending usually turns up three chapters before you know it matters. I have been reading rooms that way ever since, watching for the small thing the big thing will eventually depend on.

Automobile engineering → *Off The Pace.*

Long before the dashboards, I wanted to know what made one car quicker than the next. 'Is it the driver or the car?' is the oldest argument in motorsport and the most contested. Off The Pace is a Fabric pipeline that tries to settle it mathematically, by isolating skill from machine. (Spoiler: it's mostly the car. Usually.)

Hardgroove techno → *Systems thinking.*

Mixing 130 bpm patterns in real time turns out to be the same instinct as building real-time pipelines. Different medium, same wiring. Both reward the people who feel the structure before they hear it.

Hosting friends → *Product design.*

Cooking the food, mixing the drinks, getting the music right. It is the same instinct that builds dashboards people actually open. The only question that matters is whether anyone wants to come back. Care about that and the rest is detail.

An inability to ignore detail → *this page.*

The kerning on a heading, the easing on a transition, the exact gap between two cards. I notice all of it, which is occasionally a gift and frequently a tax. This page is set in two typefaces I argued with myself about for longer than I'll admit. I won't pretend that's healthy. I will say it's why the work tends to look the way it does.

§ III

The Pattern.

- on noticing · written from a café in dubai -

A few people have asked me how I work. The honest answer is that I notice things. Most of them don't matter. Some of them do. This is a recent one I haven't been able to stop thinking about.

FIELD NOTES
Tuesday afternoon. Specialty coffee place near JLT. Two adjacent tables, four strangers, one observer trying to mind his own business.

I was at the café with EDA. He was talking. I was supposed to be listening. Instead I was watching the two strangers at the table behind him.

Both were having what looked like completely different conversations. One was tense. One was relaxed. But the rhythm underneath them was identical. One person speaking in long uninterrupted bursts while the other touched their glass with their fingertips. Not sips. *Touches.* Fingertips to the glass, then back down, every time their partner paused for breath.

By the third repetition I realised both listeners were using the glass as a timing mechanism. A turn-taking signal. A way to regulate interruption anxiety without actually interrupting. Two people, two conversations, two completely different reasons for tension. And the same micro-gesture underneath, doing the same psychological work.

EDA hadn't noticed. Nobody at either table had noticed. The conversations themselves were louder than the pattern underneath them, so the pattern stayed invisible. It always does. That's what makes it interesting. And, to be fair to EDA, that's also what makes it a slightly antisocial trait to bring to coffee.

> *"The pattern is rarely where the noise is. This is how I think about data, too."*

Data work, done well, is mostly this. A dashboard tells you the big number went up. The interesting question is which small gesture, three weeks earlier, made the big number inevitable. Find that gesture and you can predict the next big number. Miss it and you'll spend your career reporting on weather you can't forecast.

§ IV

Off the Clock.

- what fills the rest of the day -

Outside the pipeline. Loosely ordered by how much of the week each one eats.

01 · sound

Hardgroove, bouncy, hypnotic techno.

Used to play monthly. House parties were weekly during the London years, our place, the uni gang, then their friends, then their friends' friends. No DJ name. Don't ask, just hand me the decks. Watch what 130 bpm does to a room of strangers.

02 · kitchen

Cooking, baking, hosting.

Most days. Indian primarily, but the kitchen is the kitchen. Anything that needs salt, fat, acid, and patience. The smoothies and the cocktails are why people keep coming back. I don't sell anything. I'd rather feed you.

03 · body

Push, pull, legs, arms. Four days a week.

I started lifting last year. It's the only routine that's stuck; which, if you know me and my rotations of enthusiasms, is the strongest possible endorsement. Cycling everywhere, because it's the only mode of transport that gives you back time instead of taking it. Rest is also discipline. The two rest days do as much work as the four lifting days. The body is the platform everything else runs on.

04 · craft

Crescendo, the magazine.

My amazing team and I built Crescendo together. It took half a year before everything finally came together, the way these things always go. 180 pages. GITAM University's official annual. I designed half of it, art-directed the rest, ran the marketing, designed the logo, designed the music club's logo on the side. InDesign, Illustrator, Photoshop, Lightroom, Premiere, After Effects, in the third year of engineering. I had the time because I'd learned Java in 9th grade and they were still teaching it to me in third year.

05 · attention

Procedurals about people who notice.

CHRISTIAN thinks watching procedurals is a waste of time. I think it's research. We have not resolved this. House. Suits. The Lincoln Lawyer. The Mentalist. The shows have one protagonist in common; somebody who sees what everyone else missed and then has to defend the seeing in public. Probably not a coincidence. Probably the reason I built Off The Pace, too.

§ V

A few things I had to learn.

- discovered slowly, usually the hard way -

If you came here to evaluate me and you don't read anything else, read these. They aren't clever and they aren't original. They're just the things I actually had to learn, usually slower than I'd like to admit. Maybe one of them saves you the detour.

← BACK ~

SYNC · 40MS · LIVE

TAKE DIAGNOSTIC ↗

WORKFORCE RETENTION INTELLIGENCE

ARCHETYPE *system.*

Transforming workforce sentiment into ***retention decisions***. A 13-axis diagnostic classifying employee risk across 8 behavioural archetypes - so HR leads act before attrition becomes visible.

LAUNCH DASHBOARD

TAKE THE DIAGNOSTIC

● SYSTEM_RUNTIME_LOGS

RETENTION_WINDOW: 90_DAY_PREDICTION

ENGAGEMENT INDEX	ATTRITION RISK	MOTIVATION SCORE
7.2	14%	Intrinsic

— BEHAVIOURAL SCIENCE —

THE TAXONOMY *OF INTENT.*

A 3-axis behavioural model mapping ***Engagement × Retention Risk × Motivation Driver***. 8 archetypes. Each with a distinct trajectory and intervention playbook.

● STABLE ● AT RISK ● DORMANT ● CRITICAL

01 STABLE
HIGH · SAFE · INTRINSIC
PURPOSEFUL DRIVER
The backbone. High performance, zero flight risk, intrinsically powered.

02 STABLE
HIGH · SAFE · EXTRINSIC
COMPETITIVE ANCHOR
Predictable, stable output. Reward-dependent retention.

03 AT RISK
HIGH · RISK · INTRINSIC
FRUSTRATED VISIONARY
High value but grinding against friction. Autonomy mismatch.

04 AT RISK
HIGH · RISK · EXTRINSIC
UNDERVALUED ACHIEVER
Output king with a reward gap. Retention window is closing.

05 DORMANT
LOW · SAFE · INTRINSIC
DORMANT POTENTIAL
Stable but underperforming. Environment or role mismatch.

06 DORMANT
LOW · SAFE · EXTRINSIC
QUIET COASTER
Presenteeism risk. Needs environmental or incentive shift.

07 CRITICAL
LOW · RISK · INTRINSIC
BURNING IDEALIST
Burnout trajectory. Meaning remains but energy is eroded.

08 CRITICAL
LOW · RISK · EXTRINSIC
EXIT RISK
Priority intervention. Immediate reward or structural fix needed.

— METHODOLOGICAL RIGOR —

ENGINE SCORING.

ENGAGEMENT INDEX	7.2	ATTRITION RISK	14%
Real-time psychological investment.		Exit probability within 90 days.	
MOTIVATION SCORE	Intrinsic	CONFIDENCE SCORE	88%
Categorical driver: Meaning vs. Reward.		Capability vs. goal difficulty.	
OVERLOAD INDEX	2.4	BURNOUT RISK	Low
Wait-time / output ratio. Burnout proxy.		Aggregated from Q10/Q11 sentiment.	
STABILITY INDEX	Optimal		
Manager-employee trust alignment.			

13-AXIS · DIAGNOSTIC

BEHAVIOURAL RADAR

PURPOSEFUL DRIVER

90 Days	12.4%	94.2%
WINDOW	ATTRITION RATE	ACCURACY

REALISTIC WORKPLACE

ARCHETYPE DISTRIBUTION

PURPOSEFUL DRIVER	20%
COMPETITIVE ANCHOR	18%
FRUSTRATED VISIONARY	15%
UNDERVALUED ACHIEVER	14%
DORMANT POTENTIAL	12%
QUIET COASTER	11%
BURNING IDEALIST	5%
EXIT RISK	5%

— RISK INTELLIGENCE —

THE MATRIX.

Every employee falls into one quadrant of the ***Engagement × Risk*** matrix. The dashboard surfaces the quadrant in real time - with a prescribed intervention for each.

GEMINI AI · LIVE INSIGHT

"The 'Frustrated Visionary' cohort shows a 22% increase in burnout risk correlated with workload telemetry. Recommendation: Deploy autonomy-first intervention before the 90-day window closes."

INTERVENTION LEAD TIME 90 DAYS

WINDOW BEFORE PREDICTED EXIT - DESIGNED FOR PROACTIVE RETENTION.

7-panel DASHBOARD	**onSnapshot** REALTIME
Arrow-Key PRESENTATION MODE	**CSV + TXT** EXPORT

— ETHICAL DIMENSION —

THE TRUST DIMENSION.

Most workforce tools measure performance. **This one measures trust.** The 13th axis captures how employees feel about being monitored - an ethical guardrail that prevents surveillance-driven attrition from compounding the very problem it was meant to solve.

TRUST

ETHICAL ANALYTICS

Tracking discomfort modelled as a core diagnostic metric to prevent surveillance-driven attrition.

PRESENTATION MODE

Arrow-key navigation + automated AI notes export for executive briefings.

— TECHNICAL STACK —

THE STACK.

● Firestore **onSnapshot** listeners deliver sub-40ms reactive updates to all 7 dashboard panels simultaneously.

PROJECT NAVIGATION

EXPLORE OTHER *records.*

CONTACT ● LIVE

If your data isn't driving decisions, it's just *noise.*

GET IN TOUCH ↗

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● ● ● COMPLIANCE_PROTOCOL FAIR_USE VER: 2026_EST

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F1_TELEMETRY AUDI_SPEC SPOTIFY_API

03_ FAIR_USE_REPRESENTATION

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[STATUS: ACTIVE_VERIFICATION]

>_ EXECUTE_COMPLIANCE_AUDIT.SH ⌄

← BACK ~

● PRODUCTION LIVE

VISIT SITE

SOLO ENGINEERING LEAD · LITESTORE.IN

RETAIL AS A SERVICE.

LiteStore is India's Retail-as-a-Service platform. I built litestore.in: **30+ routes**, LiteStore storefronts for **11 brands across 3 malls**, lead pipeline, GA4 telemetry. Shipped **solo at 19**.

VISIT LITESTORE.IN

● BUILD OUTPUT

```
[POST] axios → sheet.best/api/sheets/31835dc6...
```

ROUTES	TENANTS	MALLS	LOC
30+	11	3	~2K

ARCHITECTURE

HOW IT'S BUILT.

Pages Router with `getStaticProps` for every route, edge-cacheable, SEO-ready. Multi-tenant brand pages live at `/spaces/<mall>/<brand>`, each pulling per-tenant Tailwind color tokens from a single config. Forms POST via `axios → sheet.best` into the client's own Google Sheet serverless, zero backend, revocable without touching code. GA4 wired directly in `_document.js` with `process.env` config for staging/prod isolation.

NEXT.JS 12 · TAILWIND CSS · MUI · SWIPER · AOS · GA4 · SHEET.BEST · VERCEL · GITHUB ACTIONS

FORM PIPELINE

NEXT.JS FORM client-side → **AXIOS POST** sheet.best API → **GOOGLE SHEETS** client's own data

DESIGN SYSTEM

PER-TENANT TOKENS.

One `tailwind.config.js`. Every brand gets its own named color token, consumed by a shared template no copy-paste, no forked stylesheets.

● ● ● TAILWIND.CONFIG.JS · THEME.COLORS

- 🟪 **LITESTORE** #7e7ca6
- 🟧 **MENSXP** #ff5e03
- 🟩 **VITRO** #2c4b35
- 🟨 **WOW** #bc9850
- 🟧 **SLEEPYCAT** #ff6832
- 🟥 **JBL** #ff3200

ORION	GARUDA	LULU
JBL	MensXP	Frootle
Sleepycat	Vitro	
The Pant Project	Zymrat	
WOW	Skillmatics	
Xyxx		
Zymrat		
6 tenants	4 tenants	1 tenants

PRODUCTION METRICS

NUMBERS.

₹2.2 Cr+	58%	48 hrs	11	30+	3
GMV VIA FLEXI-STORES	CHEAPER THAN CONVENTIONAL RETAIL	BRAND TRANSITION TIME	BRANDS IN PRODUCTION	STATICALLY PRERENDERED ROUTES	MALL LOCATIONS

WHAT I ACTUALLY DID

MY CONTRIBUTIONS.

Shipped this as a second-year CS student. Here's the full scope code was only part of it.

SOLE FRONTEND ENGINEER

Every line of JSX, every Tailwind token, every route. Listed in /about/company as "Justin Clarke / React Developer."

BRAND IDENTITY & DESIGN

Designed the LiteStore logo and visual language from a blank brief no design team, no design file handed to me.

CLIENT LEAD (NO PM)

Sole point of contact for requirements and delivery. Directly managing a paying client at 19.

GOOGLE WORKSPACE ADMIN

Managed org-level Google Workspace for the client. First time owning infrastructure beyond the codebase.

THREE FULL REWRITES

HTML/CSS → Create React App → Next.js 12. Each version shipped. Each version taught something the next one fixed.

HIRED MY OWN REPLACEMENT

When I left, I recruited and onboarded the next developer. First time architecting a handoff, not just a product.

PROJECT NAVIGATION

EXPLORE OTHER *records.*

CONTACT

● LIVE

If your data isn't driving decisions, it's just *noise.*

GET IN TOUCH ↗

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REGION 7 | MODULE: DIS-REL-PH | ● OPERATIONAL

RELIEF LOGISTICS ARCHITECTURE

DISASTER *response.*

Modeled life-saving coordination for Philippine relief operations using an 11-entity MySQL relational engine.

MYSQL · D3.JS V7 · VANILLA JS · FORCE GRAPH · CTE + WINDOW FUNCTIONS

OPEN COMMAND DASHBOARD **PROJECT SPECIFICATION**

SCROLL

5	7	5	10	15	9	10
DISASTER	REGION	AGENCY	RESPONSE_TEAM	VOLUNTEER	SHELTER	SUPPLY

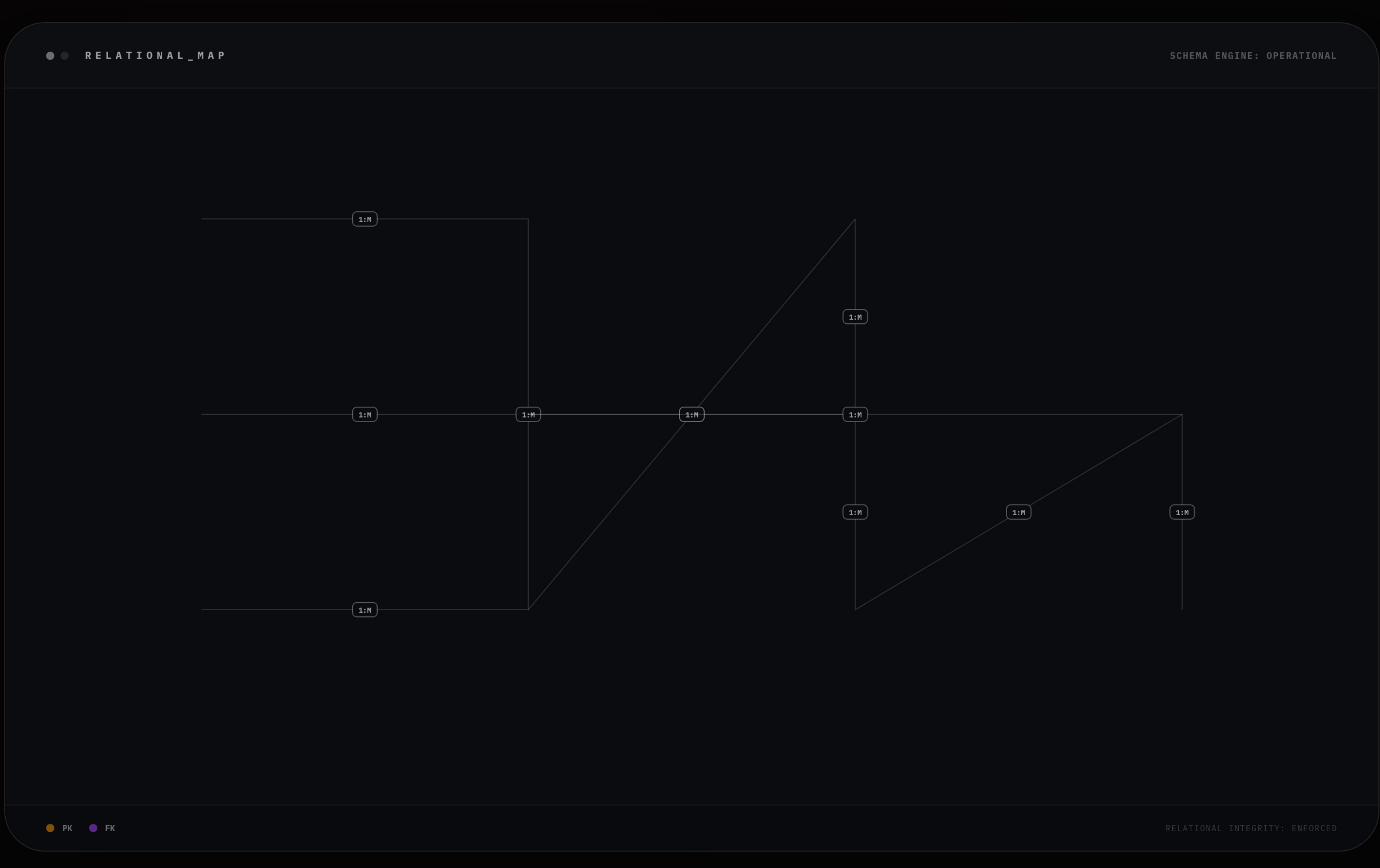
SYSTEM ARCHITECTURE

RELATIONAL BLUEPRINT.

A code-implemented visualization of the 11-entity relational engine. Toggle entity layers and explore supply chain and personnel dispatch hierarchies.

● DISASTER ● REGION ● AGENCY ● RESPONSE_TEAM ● VOLUNTEER ● SHELTER ● SUPPLY

● ● RELATIONAL_MAP SCHEMA ENGINE: OPERATIONAL



RELATIONAL INTEGRITY: ENFORCED

11	PH	M:N	11
ENTITIES	CONTEXT	RELATIONSHIPS	SYLLABUS
Core + Junction	Disaster Ops	Composite Keys	Query Categories

DATA MODELING

SCHEMA RIGOR.

Designed a robust MySQL architecture to synchronize relief efforts across 7 Philippine regions. Modeled multi-tier disasters - Typhoon Yagi, Luzon Earthquake, Mt. Kanlaon - and their full resource allocation chains.

7	47	<200ms
REGIONS	QUERIES	AVG LATENCY

ENTITY_REGISTRY_V1.0

● M:N MODELING

4 junction tables with composite Primary Keys enforcing relational integrity across the volunteer-to-team and supply-to-region chains.

● CHECK INVARIANTS

CHECK constraints enforce current_occupancy ≤ capacity across all 9 shelter nodes - prevents logical data violations at the schema level.

● FK INTEGRITY

Foreign Key chains span regional dispatch zones: Agency → Team → Disaster → Region → Shelter, ensuring cascading constraint coverage.

● GEOGRAPHIC OPS

NDRRMC and PRC agency coordination nodes model real Philippine relief hierarchy across 7 affected regions and 5 concurrent disaster events.

VISUALIZATION ENGINE

THE D3 ENGINE.

The interactive dashboard is 1,603 lines of handwritten Vanilla JS - no React, no framework. Built on D3 v7 force simulation with custom physics, multi-layer filtering, and a programmatic alert system.

1,603	61	71	4
LINES OF CODE	GRAPH NODES	EDGES	ALERT TIERS
Vanilla JS · no framework	D3 v7 force simulation	6 relationship types	Critical · Warning · Info

◉ **FORCE SIMULATION**

D3 force-directed graph with custom charge, link distance, and collision physics tuned for 61-node readability.

◈ **PROGRAMMATIC ALERTS**

Alert panel generated dynamically from node data - shelter load thresholds, active disaster states, standby team detection.

◎ **MULTI-LAYER FILTERING**

Entity type toggling, full-text node search, edge hover inspection, pin/unpin state, and keyboard controls.

ANALYTICAL BREADTH

THE QUERY SYLLABUS.

● ● ● SHELTER_STRESS_TIER.SQL

```sql
WITH ShelterMetrics AS (
  SELECT name, capacity,
    (occupancy / capacity) * 100 AS stress_pct
  FROM Shelters
)
SELECT name, stress_pct,
  NTILE(3) OVER (ORDER BY stress_pct DESC) AS stress_tier
FROM ShelterMetrics;

-- Identifies high-risk evacuation zones
-- Shelter 8 Bicol: 99% · Shelter 1 Manila: 96%
```

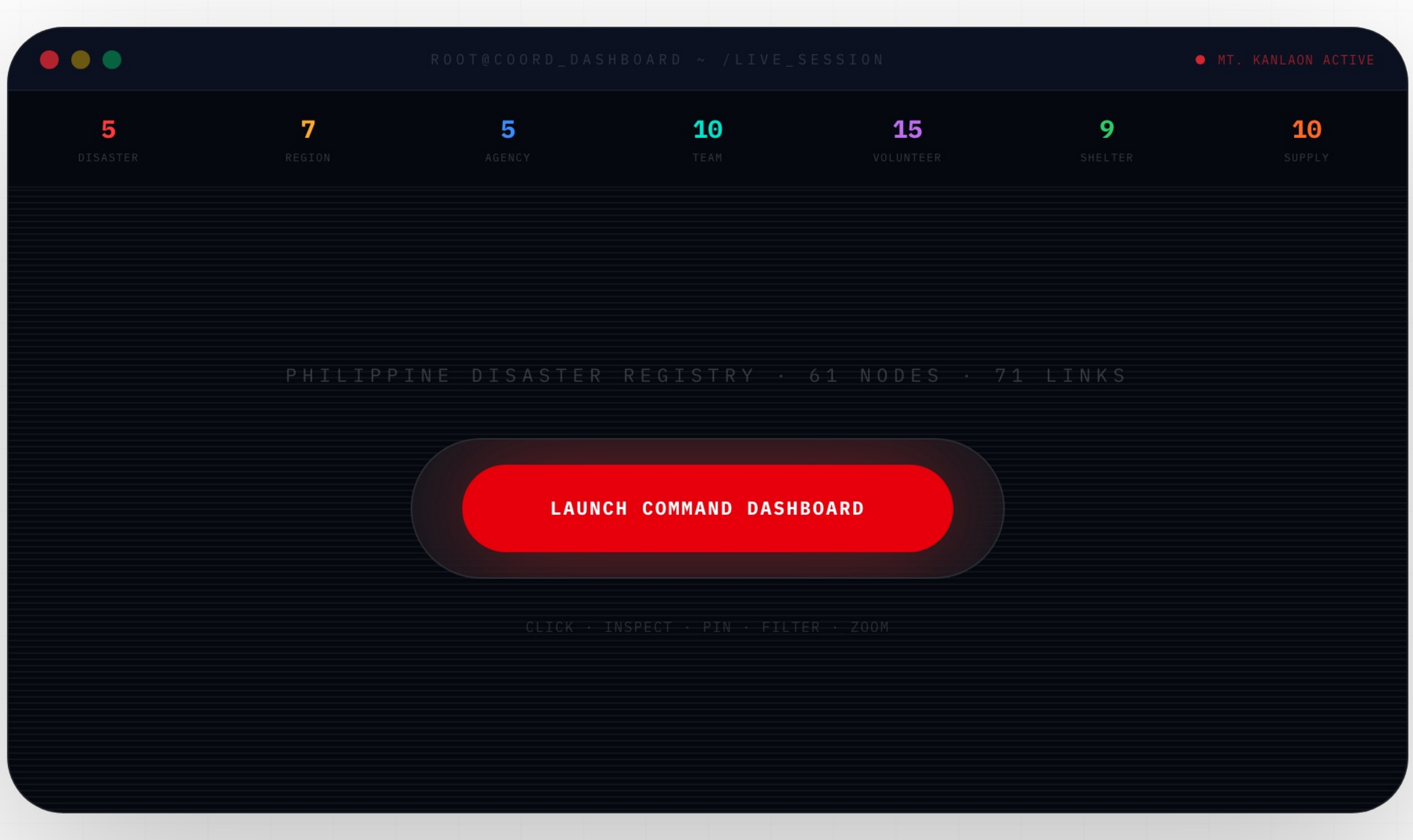
11 query categories covering the full analytical spectrum - from basic retrieval to window functions, CTEs, and composite key modeling. Each category applied to real Philippine disaster data.

01 SELECT/WHERE/ORDER
03 GROUP BY + HAVING
05 RANK / DENSE_RANK
07 IN-SUBQUERIES
09 COMPOSITE PKS
11 JUNCTION MODELING
02 INNER/LEFT/RIGHT JOIN
04 CTE COMPOSITION
06 NTILE BUCKETING
08 TIMESTAMPDIFF
10 CHECK CONSTRAINTS

INTERACTIVE COMMAND INTERFACE

THE DASHBOARD.

7 color-coded entity types. Full-text node search. Edge inspection on hover. Programmatic alert system. Pin/unpin. Zoom. Reset. All in a single HTML file.



ROOT@COORD_DASHBOARD ~ /LIVE_SESSION ● MT. KANLAON ACTIVE

5	7	5	10	15	9	10
DISASTER	REGION	AGENCY	TEAM	VOLUNTEER	SHELTER	SUPPLY

PHILIPPINE DISASTER REGISTRY · 61 NODES · 71 LINKS

LAUNCH COMMAND DASHBOARD

CLICK · INSPECT · PIN · FILTER · ZOOM

11	47	1,603	<200ms
ENTITIES	QUERIES	LINES OF CODE	AVG QUERY LATENCY
core + junction tables	11 analytical categories	vanilla JS, no framework	indexed relational design

PROJECT NAVIGATION

EXPLORE OTHER *records.*

CONTACT ● LIVE

If your data isn't driving decisions, it's just *noise.*

GET IN TOUCH ↗

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← BACK ~

SP-442 | ● PRODUCTION READY

MSC RESEARCH · SPOTIFY PROJECT

SPOTIFY: *predictive* ENGINE.

A **Terminal-First** research artifact solving the cold-start problem through 3-dimensional hierarchical similarity and acoustic DNA modelling.

READ PROJECT REPORT

VIEW NOTEBOOKS

ACOUSTIC FEATURE VECTOR

ACOUSTIC DNA.

Each track is fingerprinted across 20+ Spotify API dimensions. The **green** features drive high-energy recommendations; **teal** features fine-tune acoustic texture. Together they form the similarity vector.

DANCEABILITY	82
ENERGY	74
VALENCE	91
ACOUSTICNESS	12
INSTRUMENTALNESS	5
LIVENESS	18

SAMPLE VALUES · NORMALISED 0–100

SIMILARITY_ENGINE.PY

```
# Build audio feature vectors
audio_vecs = scaler.fit_transform(df[AUDIO_COLS])

# TF-IDF encode genre strings
genre_tfidf = TfidfVectorizer().fit_transform(df['genres'])

# Compute cosine similarities
sim  = cosine_similarity(audio_vecs, query_audio)
sim2 = cosine_similarity(pop_vecs,   query_pop)
sim3 = cosine_similarity(genre_tfidf, query_genre)

# Hierarchical stable sort → cold-start resolved
df['sim'], df['sim2'], df['sim3'] = sim, sim2, sim3
result = df.sort_values(['sim3','sim2','sim'], ascending=False, kind='stable')

```

METHODOLOGICAL DISCIPLINE

THREE MODEL *BENCHMARK.*

We tested our engineered models against Spotify's internal baseline - measuring precision across three distinct input modes: Playlist, Song, and Artist. Only by competing against the source can you beat it.

RECOMMENDED
MODEL 1
01

HIERARCHICAL

The flagship model. Uses a stable sort across 3 similarity dimensions: Genre TF-IDF → Popularity (Artist/Track) → Acoustic DNA.

SORT PIPELINE

GENRE TF-IDF → POPULARITY → ACOUSTIC DNA

BALANCED
MODEL 2
02

AVERAGED HYBRID

A linear combination of vector distances. Balances genre, popularity, and acoustic features into a single similarity score.

SORT PIPELINE

GENRE → POPULARITY → ACOUSTICS

CONTROL
CONTROL
03

SPOTIFY BASELINE

The official Spotify API recommendations endpoint - our external benchmark for measuring model performance.

SORT PIPELINE

SPOTIFY API → INTERNAL → BLACK-BOX

MEMORY-FIRST DESIGN

DATA ENGINEERING.

Handling a 1M-row dataset on edge environments required strict memory engineering. We utilized **float16/int8** downcasting and chunked CSV processing to maintain a sub-200MB runtime footprint.

DATASET **1M Rows** Spotify MSD	MEMORY **<200MB** float16/int8
INGESTION **Live API** Spotipy + cron	FEATURES **20+** Audio + Meta

DATASET GROWTH LOOP

01 **PLAYLIST URL INPUT**
User pastes Spotify playlist URI

02 **SPOTIPY API FETCH**
Retrieve audio features for all tracks

03 **APPEND TO DATASET**
New tracks persisted to new_tracks.csv

04 **TRIGGER AT 200 ROWS**
Auto-merge into main 1M corpus

NOTEBOOK REPOSITORY

RESEARCH ARTIFACTS.

STAGE 01	**Preprocessing.ipynb** Data cleaning, float16 downcasting, and CSV chunking logic for sub-200MB edge runtime.	312 LINES .IPYNB	→
STAGE 02	**Feature_Extraction.ipynb** Spotipy API ingestion, 1M-row MSD parsing, and multi-threaded feature engineering.	487 LINES .IPYNB	→
STAGE 03	**Modeling.ipynb** Hierarchical sort validation, cosine similarity benchmarks, and Model 1 vs. Spotify Control evaluation.	604 LINES .IPYNB	→

CORE: React + TypeScript   AI: Gemini 1.5 + Scikit-learn   DB: Firestore (real-time)   INPUT: Playlist · Song · Artist

PROJECT NAVIGATION

EXPLORE OTHER *records.*

CONTACT

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[STATUS: ACTIVE_VERIFICATION]

>_ EXECUTE_COMPLIANCE_AUDIT.SH ⌄

JUSTINCLARKE@PORTFOLIO: ~

email linkedin github | Royel Otis - Linger - SiriusXM Session | ● shipping

```
// session_active: port 8080 // auth_success: 13:56:50

~$ whoami
```

justin *clarke.*

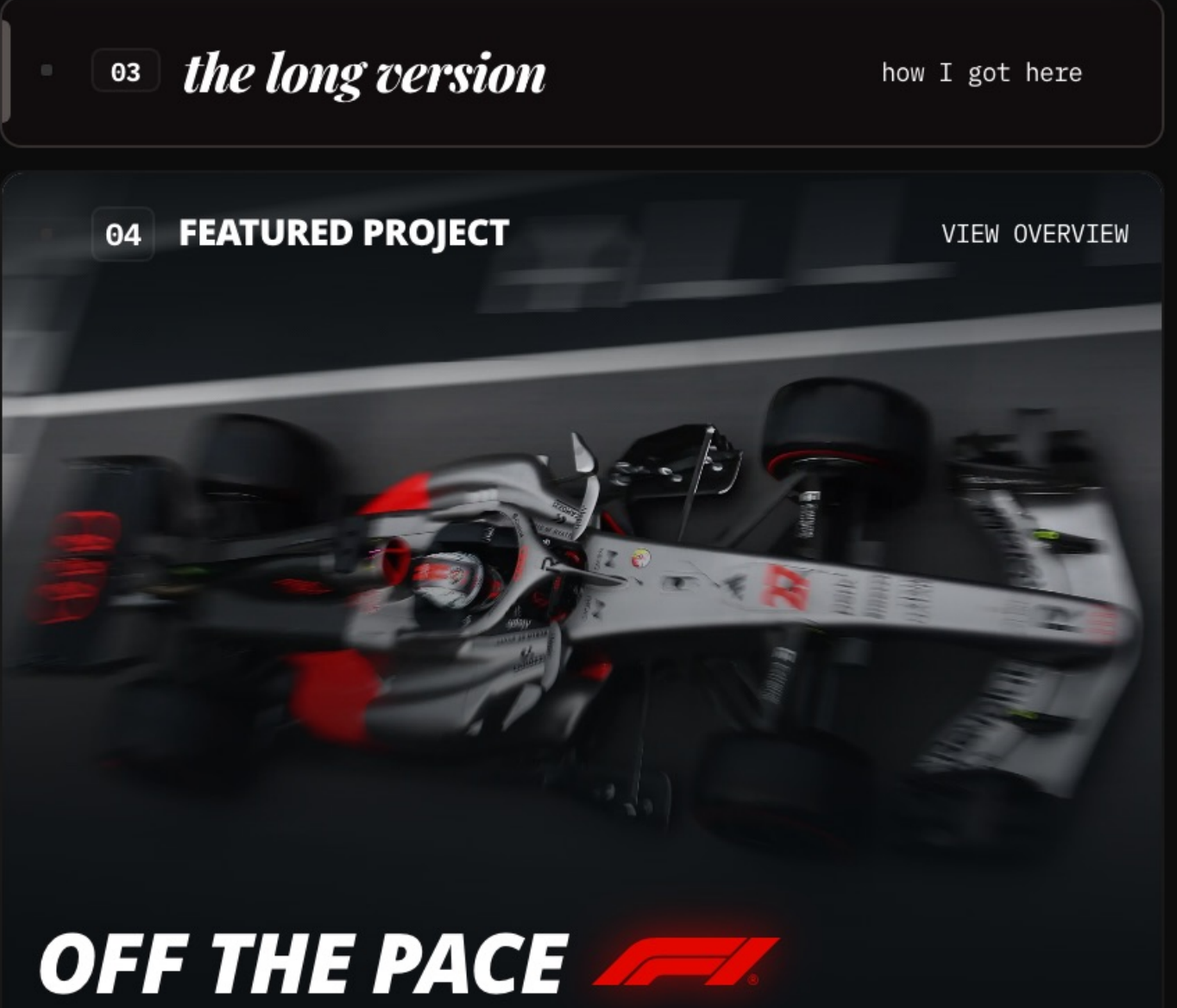
```
~$ profile.read()
```

data & analytics developer.

Data to decisions and decisions into products

SYS CONTROLS 04 CH

- 01 **RESUMÉ** DOWNLOAD PDF
- 02 **CONNECT** GET IN TOUCH
- 03 *the long version* how I got here
- 04 **FEATURED PROJECT** VIEW OVERVIEW



OFF THE PACE

2.4M	dbt	XGBoost
TELEMETRY ROWS	TRANSFORMATIONS	PACE MODEL

MANIFEST

Featured *projects.*

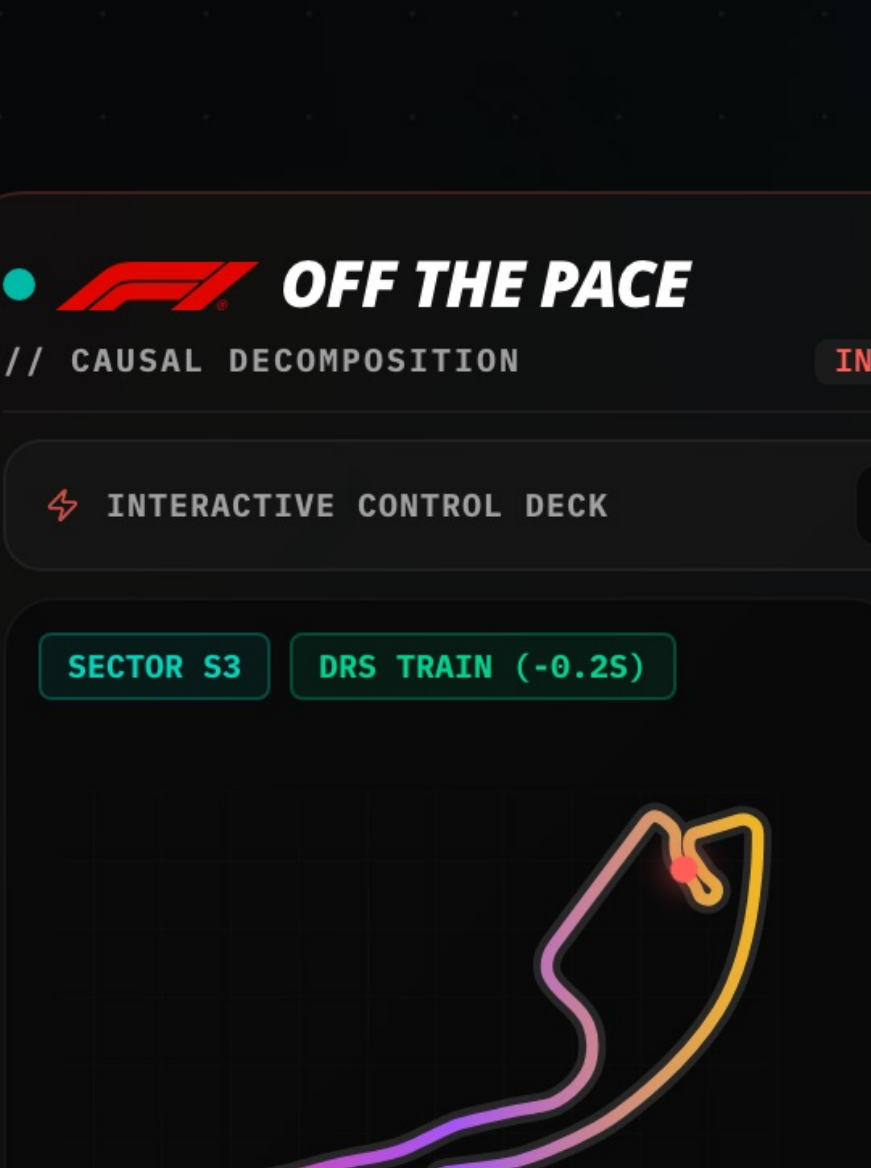
PORTFOLIO DECK // CLICK TO EXPLORE WORK

OFF THE PACE

// CAUSAL DECOMPOSITION INGESTING • LAP 4/57 AGE: 5 LAPS

INTERACTIVE CONTROL DECK TELEMETRY CAUSAL

SECTOR S3 DRS TRAIN (-0.2S)



S1 28.4s · S2 36.1s · S3 20.8s
SURF: 94°C · BULK: 89.7°C

Monaco Spa Monza

```
(E)        79.408s
           +0.600s
g):        +3.798s
           -0.200s
:          +0.060s
FF E:      -1.350s
```

dbt CONSOLE RUN TESTS

```
[13:49:12] [dbt] SUCCESS…
[13:49:13] [XGBoost] Run…
[13:49:14] [Fabric] Sync…
[1:56:47 PM] [DuckDB] In…
[1:56:48 PM] [dbt] SUCCE…
```

2.4M Telemetry Rows · Fabric + dbt + DuckDB · Causal Regression Model

LET'S WORK TOGETHER VISA · SPONSORED →

LITESTORE

PROD SAAS · PROD HEALTHY · LCP: 0.6S

VERCEL EDGE · NEXT.JS LIGHTHOUSE TELEMETRY

98 LCP	99 FID	96 CLS	94 TTFB	95 INP

LIGHTHOUSE AUDIT

Performance	98
Accessibility	92
Best Practices	95
SEO	100

0.6s LCP · 80% Faster · +20.4% CVR · GA4 + Vercel

DISASTER RESPONSE

RELATIONAL DATABASE · SCHEMA OPERATIONAL · DISASTER: 5

DIS-REL-PH · MODULE SCHEMA QUERIES

5 · 7 · 5 · 10 · 15 · 9 · 10

SCHEMA FEATURE

M:N Rel Modeling
4 junction tables · composite PK setup

11 Entities · 47 Queries · P95 < 200ms · FK Integrity

SPOTIFY ENGINE

MSC RESEARCH · MODEL OPERATIONAL · VALENCE: 0.91

MSD-MSC · 12D SPACE VECTORS RECOMMEND

91 VAL	65 TEM	74 ENE	82 DAN	12 ACO	18 LIV

MODEL FEATURE

TF-IDF Genre Vector
Sparse term weighting across 4.2K labels

1.2M Tracks · 12D Vector · TF-IDF + Cosine · MSD

HR ANALYSIS

BEHAVIOURAL AI · FLIGHT RISK DETECTION · 8 ARCHETYPES

COLLECT → MAP → ANALYZE → GENERATE

TRACKED METRICS 7 dimensions · AVG ENGAGEMENT 5.4/10

8 high performers show workload misalignment.

127 Analyzed · 8 Archetypes · Engagement scoring · Risk modeling

EXPERTISE

What I bring to the *stack.*

HOVER SKILLS TO TRACE PIPELINE

LINEAGE CONSOLE | Awaiting node connection. Hover any skill below to trace pipeline lineage. NODE: none / STATUS: STANDBY

01 STAGE 1
ENGINE & LOGIC
DEEP QUERY LOGIC & INFRASTRUCTURE

- SQL
- Python
- dbt
- Medallion Architecture
- DuckDB-Wasm

02 STAGE 2
SIGNALS & BI
PREDICTIVE SIGNALS & TELEMETRY

- Power BI
- Microsoft Fabric
- Causal Inference
- Telemetry Platforms
- XGBoost / ONNX

03 STAGE 3
PRODUCT & GROWTH
PRODUCT ANALYTICS & APPLICATIONS

- React
- KPI Development
- Event Tracking
- Data Products
- A/B Testing

CAREER TIMELINE

The full *record.*

SELECT CARDS TO EXPAND DETAILS

2018 · 2020 · 2022 · 2024 · 2026 NOW · 2028

PROFESSIONAL

JAN 2024 – DEC 2025
VNS Solutions
ANALYTICS ENGINEER

- ↓ **12+ hrs/wk** MANUAL WORK AUTOMATED
- **End-to-end** ANALYTICS LOOP OWNERSHIP

MICROSOFT FABRIC · POWER BI · SQL · PYTHON

APR 2021 – APR 2023 ↔ BTECH → MSC
LiteStore
TECHNICAL LEAD, FULL STACK & WEB PERFORMANCE

- ↑ **20%** CONVERSION VIA A/B TESTS
- **3.0s → 0.6s** PAGE LOAD (SSR + CACHING)

NEXT.JS · AWS · VERCEL · GA4

JAN – APR 2021 ↔ DURING BTECH
Drop
FRONTEND DEVELOPER

REACT · TYPESCRIPT

ACADEMIC

FEB 2026 – 2028 ● ACTIVE
Birla Institute of Technology and Science, Pilani
MBA IN BUSINESS ANALYTICS

ANALYTICS · STRATEGY · ML

— Intersection of data analytics and business strategy. Evening programme - fully available for full-time roles.

2022 – 2023
Queen Mary, University of London
MSC IN COMPUTER SCIENCE

PYTHON · ML · RESEARCH

2018 – 2022 DISTINCTION
Gandhi Institute of Technology and Management
BTECH, COMPUTER SCIENCE & ENGINEERING

C/C++ · ALGORITHMS · MATHS

CERTIFICATION IN PROGRESS

Microsoft PL-300: Power BI Data Analyst ● EXPECTED JUN 2026

CONTACT ● LIVE

If your data isn't driving decisions, it's just *noise.*

GET IN TOUCH ↗

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SOURCE_MANIFEST
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